

Proposal for Greenfield Local hub (GLH)

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1. Research Appendix

How is hardware and software used in the retail (food and produce) sector?

Software

There's lots of different software used in retail. For example:

- point of sale (POS) software – tills, ensures safe transactions
- retail inventory management software – allows staff to check stock of products to restock, generate data on best selling products etc
- cloud-based accounting applications – allows remote monitoring of profits and outgoing to ensure all is properly paid for and all money is accounted for
- customer relationship management (CRM) software – allows for customers to be able to contact companies for any issues to build trust and professional relationships with customers.

Source: <https://www.unleashedsoftware.com/blog/retail-software/>

Hardware

Necessary (POS) hardware:

- POS terminal (till)
- Barcode scanner

- receipt printer
- cash drawer
- payment terminal

source: <https://retailsolutions.ie/blog/5-necessary-point-of-sale-hardware-components/>

Newly emerging technologies within the retail industry

Future technologies of retail:

- interactive shop windows
- self-service kiosks
- transparent touchscreen displays
- NFC/RFID tagged smart products
- touchscreen tables with object recognition
- XXL Touchscreens with multi-user Technology
- customisable apps for large-scale touchscreens
- facial recognition as an analytics tool
- smartphone integration for multi-channel marketing

source: <https://www.eyefactive.com/en/whitepaper/future-of-retail-9-key-technologies>

Retail and food regulations

Food standards act 1999

Gives power to act in the consumers interest at any stage in food production and supply chain.

This ensures that food is made to a good sell-able standard to keep the consumer safe. This means that food does not include, remove anything from food or treat food in any way that means it would be damaging the health of those eating it. Food being sold is of the quality consumers would expect and food is labelled, advertised and presented in a way that is not false or misleading.

General food law

There is mainly legislation into protecting:

- food imports and exports
- safety of products and staff
- traceability of products
- proper labelling

- product withdrawals and recalls

source: <https://www.food.gov.uk/about-us/key-regulations>

2. Organisation Overview

Greenfield Local Hub (GLH) is a co-operative of local farmers and food producers that runs and manages a retail outlet for locally produced food and drink, Collectively market products and manage stock and product traceability.

They would like a solution that allows a customer to receive information about the producers/suppliers about the products they are buying, place orders for collection delivery and manage their account and order history.

3. Problem Overview

Currently, GLH doesn't have a suitable digital system that allows customers to easily order products online. This reduces the range of customers they have as not all customers are able to go to their retail outlet regularly and may choose to use big brand supermarkets instead.

This means that profits are not being made and products are not being used, meaning they are being wasted and disposed by GLH. This means that they are losing potential income and having to pay to dispose of expired products as to adhere to food regulations.

The current system requires customers to travel to the retail outlet and purchase things in store. Not all customers have time/ are too far away/ are incapable etc to purchase their produce in person. A digital system would alleviate this problem and allow a wider range of people to be able to purchase this product as they will be able to get it delivered or pre-order products for pickup.

4. Scope

For the scope of this project, I will be focusing on the user order system that allows them to place orders, manage their accounts and view products. This will ensure that it is built to the highest quality and standards available with the time constraints I have.

This means, that I will be missing out on other important factors for building an order system. e.g financials (payments, storing card details securely etc), marketing the system, authenticating logins via a text, email etc.

In a Real Life scenario the project would have a team of people working on the solution. This means that the scope is realistic as different people would focus on the areas of code that they are best at producing. This ensures that the code is produced to the highest quality and in a timely manner to fit a pre-decided time frame that would be agreed upon with the client.

5. User Groups

The GLH system will have a range of users, most likely by people 18 - 65 years old who need produce and who can navigate the web to order products. Customers will require different quantities of products to fit their needs (eg a family, single household, restaurant etc). The system will need to be simple and clear to navigate and allow users to know what they are ordering.

Staff using the system will be using the system everyday to manage the products and quantity available, orders to pack/ deliver etc.

Customers

Customers may only use the system a couple of times to regularly to order their weekly shop. Depending on how easy the system is to use and if the products fit their needs. This means that the system looking appealing and clearly displaying their products, prices etc is crucial to help retain repeat customers.

Families/ households

Households will use the system to arrange the products that they regularly use day to day (the weekly shop) to be ordered to their home or collected. Customers will be a wide range of ages as stated above, be ordering products for an average of 2-6 people and have a range of technological abilities (very weak skills to highly skilled).

General consumers like households are more likely to be enticed and make use of loyalty schemes and points. Though are less likely to repeatedly order the exact same thing repeatedly as a business might.

Offices / other businesses – Businesses

Businesses will use the system to order products for usage of their office workers. These businesses will be ordering a range of produce to fit the needs of their employees (eg milk, fruit etc) The business will have an employee manage this and keep the deliveries on a regular schedule. This means they will want a set amount of produce at a set time every week to every fortnight.

It is important to consider the difference in a general consumer and a business as a consumer for loyalty schemes, order management etc as they are more likely to want features like set repeat ordering rather than reward points.

Restaurants / Pubs / cafes – Businesses

A Business that is using the products they are using to generate profits are going to be the customers purchasing the largest amounts of products compared to other business and general consumers. This is because they will be selling products made with the purchased produce.

This means that they will need a large quantity of food on a regular basis. They will value, consistency in produce, high quality produce and proper time management so that they can organise stock. They will be repeat customers as they will only use a handful of food producers to stock their ingredients.

These customers are most likely to pay a higher price than other consumers from other businesses/general consumers, but since they are purchasing stock in bulk they may require a lower price per individual product than other consumers. This should be considered when developing the system.

Staff

Staff will be using a staff version of the system for a vastly different purpose than customers. They will need to add products, update stock of products, manage and deliver orders etc. Different staff members will need to be able to see different things on the system dependant on their role to protect the data of customers under the data protection act 2018 (see 11)

Delivery drivers

Delivery drivers will need to see the address of customers they are delivering products too. It is important that they only see the required data needed for each customer to protect their data.

They will also need the system to work on a mobile device so that they can deliver multiple orders in one trip and a GPS system to plan their delivery routes so that less time is wasted on deliveries. Keeping products fresh and jobs done efficiently and easily for the drivers.

A new system that can incorporate that for staff will increase the happiness of delivery drivers, making them work happily and efficiently. Improving their work experience and conditions as well as improve customer satisfaction in the delivery time of products.

Admin

Admins will be in charge of making accounts, managing data, stock management etc. This means they will need to be able to make staff accounts and set their permissions based on what they need to do their job correctly. Making the system easier for employees to use, whilst also protecting the data of all users.

Farmers/ Suppliers

Farmers and suppliers will be providing the products that the system sells. It is important that it is made clear what products come from where to meet up with food regulations (see 1). They will need to ensure the details of their products are truthful and are displayed to meet their needs. Eg a nice picture of the product, a good description etc.

They may not be interacting with the system directly (eg adding stock, managing accounts etc) they play a crucial role in the systems success.

Store Staff

Store staff are the users who will be managing the products on the system. They will have to determine how many of each product will be able to purchase online vs instore and how they want both to remain meaningfully stocked (eg. having all stock available to purchase on both, having a set amount on the online system etc)

They will need to be able to manage stock on the system and see users orders to pack them and properly organise them to be sent off for delivery or be collected in store.

6. Empathy Maps

Families/ Households

Says I want my family to eat healthier food I want my food to be sourced locally I don't have time to go out of my way to go to the farm shop	Thinks food from a big supermarket will have to do locally sourced food would be ideal but it is out of my way to go all the way to the farm shop. I prefer that my food is delivered because its easier for me.
Does Orders food from a large supermarket instead of locally. Has to wait in long queues and fight for delivery slots. Deal with auto generated substitutions	Feels annoyed trying to get a slot for delivery of their weekly shop disappointed by food substitutions not meeting their needs. Having to pay higher prices for lower quality products with a large amount of food miles. Disappointed in what they have to feed themselves and their family.

Restaurants / Pubs / cafes – Businesses

Says I need a supplier that will meet my needs for delivery, product quality and professional relationship. I want to be able to see where my food is sourced so I can inform customers I want to supply as many ingredients locally where I can. I need to be able to order food in bulk, regularly I cannot accept food substitutions	Thinks I am struggling to find a good supplier that offer me a range of products. It is hard to keep track of a lot of different suppliers for different things. I need to see what is in ingredients so I can easily find allergens
Does Looks online to find different suppliers	Feels Stressed about finding proper food suppliers who

researches where their food is sourced and what is in it Has to insure that all allergens in ingredient match the menu. Find alternative products for allergens. Has to build a relationship with suppliers to ensure their needs are heard and met.	can meet their needs worried about deliveries not coming at the agreed time and being left without certain ingredients to make food to serve. Confused from having to deal with different suppliers for different ingredients. Annoyed at the lack of clear communication on what is being sold and what is in it.
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Delivery drivers

Says I need to be able to see my customers order numbers and addresses so I deliver the correct orders to the correct people I need to be able to access the system remotely so that I can see where I am going and what's next. I need the routes to be preplanned so reduce time wasted on the road	Thinks I don't know where I am going for certain orders. It is hard having to plan my own routes and it is wasting time I keep getting stuck in traffic because I cant predict it in live time. It is hard to locate customers addresses without a proper GPS system
Does Has to plan their own routes, which isn't as efficient as a pre-planned one gets stuck in traffic because they don't know its there. Is late and messes up the delivery of orders.	Feels stressed on meeting delivery quotas. Confused on what order goes to each customer frustrated at getting stuck in traffic and not knowing how to avoid it.

Farmers/ Suppliers

Says I want a way for a wider range of customers to be able to purchase my products. I want customers to be able to find more information on how I produce their food/ products. I don't want my products expiring and being wasted	Thinks Too much of my produce is expiring and not being sold so I am losing money. I would like to be able to see the quantity of what I have put to be sold at GLH so I can manage my stock. I would like to be able to see which products customers like so I can produce more.
Does Has to calculate their own stock and what is being sold at GLH. Only selling products in person which means produce is not being sold and getting wasted Only a small range of customers are available to visit in person and purchase products.	Feels anxious about low sales and high amounts of waste stressed about calculating stock numbers in case of accidental mistakes unsure on what customers are demanding vs what isnt selling very well.

7. User stories

Families/ Households

As a/ an... (role)	Household
I want.. (goal)	To be able to source good quality food to consume in my household. Be able to order locally produced food to my household. To be able to get food for cheaper and of a higher quality than super markets.
So that.. (benefit)	So that my family can eat better food, produced locally without having to travel a long distance and at a reasonable price.
Acceptance Criteria (conditions of satisfaction)	
• Be able to see what products are available and see their ingredients • be able to select a delivery date and time • clearly see costs and where their order is • be able to contact someone easily if there are any issues • need to be able to easily understand and order food • a loyalty scheme that offers users discounts and special offers	

Offices / other businesses – Businesses

As a/ an... (role)	office
I want.. (goal)	To be able to supply my workers with fruit, milk and other produce to improve working conditions
So that.. (benefit)	My employees are happy and have access to snack and food whilst working
Acceptance Criteria (conditions of satisfaction)	
• Be able to have an automated order on a set time frame so they don't have to reorder the same food over and over again. • Be able to have a wide variety of products each week based on what is available each week	

Admin Staff

As a/ an... (role)	admin
I want.. (goal)	To be able to make accounts with limited access so workers can only access what they need.
So that.. (benefit)	Workers are less overwhelmed using the system and user data is protected because they can only see what they need to see.

Acceptance Criteria (conditions of satisfaction)
• Be able to make accounts for workers • be able to limit access to what they need each employee to see to complete their job • protect user data to comply with GDPR (see 11)

Store Staff

As a/ an... (role)	Store Employee
I want.. (goal)	To be able to manage the stock levels of different products and pack online orders for collection or delivery.
So that.. (benefit)	I can manage what stock is available to purchase in person and online and ensure that people receive the correct orders.
Acceptance Criteria (conditions of satisfaction)	
• Be able to view and alter stock levels to ensure it is updated correctly • be able to view customer orders for packing and sending off for delivery/ collection • be able to verify that collection order are given to the correct customers • reduce the amount of miscalculations of stock • reduce customers accidentally receiving incorrect orders.	

8. User acceptance Criteria

no.	Criteria	How generated	Why important
1.1	Clear navigation bar	Research through alternative systems (see 12)	It is important so that users can easily understand and navigate the site.
1.2	Be able to see what products are available and see their ingredients	restaurant empathy maps (see 6)	To check allergens in products, following legislation on food regulations (see 1)
1.3	clear instructions on each page	following law (see 11) and alternative system research (see 12)	to make it easy for users to understand and reduce stress when using the system
1.4	fast loading times (2-5 seconds max)	Research through alternative systems (see 12), NFR (see 9)	fast loading times are needed to retain customer attention and to ensure the website looks professional
1.5	alter orders	user stories (see 7)	customers may change their mind on what they need to purchase and will want to have the ability to alter their order before it is delivered. Users need to be able to see what

1.6	view orders	user groups (see 5)	they have ordered and ordered in the past so they can keep track of spending, what works for them, what doesn't etc.
1.7	be able to select a delivery date and time	user stories (see 7)	users need to be able to select a time that works for them for food to be delivered so that they can be there when it arrives.
1.8	clearly see costs	Research through alternative systems (see 12)	Users need to be able to see how much products cost so that they can plan and buy products that match their budget
1.9	be able to contact customer support easily	user stories (see 7)	Users need to be able to contact customer support to report any issues they may have. Customers like good customer support that isn't frustrating to work with.
2.1	automated orders	user stories (see 7), empathy maps (see 6)	Users may want the same food delivered on a set schedule (eg businesses). Automated repeat ordering can make this easier on them so they don't have to waste time placing orders every time.
2.2	product organisation	user stories (see 7), Research through alternative systems (see 12)	Products need to be organised in a meaningful way so that users can find them and know what they are purchasing.
2.3	make staff accounts	user stories (see 7)	Admin staff need to be the only accounts able to make staff accounts so that the system can't be infiltrated and for UAC 2.4
2.4	limit what staff accounts can access	user stories (see 7)	staff should only be allowed to see data related to complete their jobs in line with GDPR (see 11) to protect the data of users. It is also important to limit access to reduce confusion when using the staff system
2.5	view orders to deliver/collect	empathy maps (see 6), user stories (see 7)	staff need to be able to see who ordered what to package and deliver orders to the correct customer.

9. Functional and Non-Functional Requirements

Functional Requirements (Business Needs)

1. Customer Needs

no.	Criteria	How generated	Why important
1.1	Users should be able to register and login to the system securely	KPI's (see 10), risks (see 13)	To prove an increase in account creation (KPI 1.2, see 10), users need to be able to securely make and login to accounts. It needs to be secure to ensure that accounts are not intercepted or created by bots to prevent Denial of service (DOS) attacks.
1.2	Users need to be able place an order	empathy maps (see 6)	Users need to be able to place an order for a product for the system to work effectively and sell products.
1.3	Users need to be able to alter and delete orders	User acceptance criteria (see 8)	Users need to be able to add or remove items from their order before it ships out to fit their needs. (families/ households may change their mind on what they want to order)
1.4	Users should be able to view their orders	user stories (see 7)	Users should be able to view their order and its details so that they know what they have ordered and where/ when its coming to check all the information is correct and to verify the order has been placed successfully.
1.5	users should be able to log out of their account	Decomposition (see 14)	Users need to be able to log out of their account so that forms aren't resubmitted (causing data to be duplicated in the database) and so accounts are kept secure.
1.6	users should be able to view their account details	law/ legislation/ guidelines	Users need to be able to see their account details so they can be kept up to date. Ensuring the data isn't redundant/ useless is important for security (passwords etc). It also

		(see 11)	helps ensure that the system is adhering to GDPR/ DPA 2018 (see 11).
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2. Staff Needs

no.	Criteria	How generated	Why important
2.1	Staff need to be able to login to the system securely	Risks (see 13)	Users need to be able to securely make and login to accounts. It needs to be secure to ensure that accounts are not intercepted and use to access data on customers (eg address) for malicious purposes.
2.2	Admin staff need to be able to create accounts and set permissions	laws (see 11), risks (see 13), User Stories (see 7)	Only admin staff being able to create staff accounts prevents random people signing up to the staff system and ensures that permissions are correctly set to fit the employees role so that they can only access what they need to for their job (e.g only accessing necessary data on customers to complete work)
2.3	Staff need to be able to view orders	empathy maps (see 6), risks (see 13), law (see 11)	Staff need to be able to view orders so that they can be packed correctly. Only staff members in charge of packing orders should be able to see the contents.
2.4	Staff need to be able to view and update stock	empathy maps (see 6), risks (see 13), law (see 11)	Staff in charge of keeping stock need to be able to see what they have received and update the system accordingly so that users can purchase it. Only users who should be updating stock should be able to, to prevent malicious misuse of the system so that the system is secure.

3. General Needs

no.	Criteria	How generated	Why important
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3.1	The website needs to be accessible and easy to navigate	Law (see 11), KPI 1.5 (see 10)	The system needs to be accessible to all users under the equality act (see 11). It is best done by following WCAG guidelines to ensure that all users are considered. It is also important to retain customers as they will not become frustrated with the system and are more likely to choose it again.
3.2	The system needs to display information about each product clearly	law (see 11), food standards (see 1), User stories (see 7)	The system needs to clearly state what a product is and its ingredients so customers are properly informed on what they are buying (protecting their right under the consumer rights act (see 11)), ensuring all products information and standards are up to date with food standards (see 1) and so customers are kept happy because they have received what they wanted when ordering.

Non-Functional Requirements (How the system should do things)

1. User Needs

no.	Criteria	How generated	Why important
1.1	System needs to be easy to use	Research through alternative systems (see 12)	The system needs to be understandable and easy to use so users choose it over alternative providers
1.2	system should be able to run on a portable device	empathy maps (see 6)	system needs to be able to run on portable devices for staff who are delivering products (remote access) and users who are using the system on a portable device (ordering via a phone) as is the current norm

2. General Needs

no.	Criteria	How generated	Why important
2.1	Passwords need to be hashed using sha256	Risks (see 13)	Passwords need to be hashed with an up to date hashing method (sha256) so that passwords are stored securely in the database
2.2	Pages need to load in 2-5 seconds max	Research through alternative systems (see 12), UAC (see 8)	Pages need need to load in a timely matter to retain customer attention and ensure the website looks professional.
2.3	system should be adaptive to change.	laws/ legislation/ guidelines (see 11), risks (see 13)	The system needs to be adaptable to changes in legislation and to meet the clients needs in the future. The system not being upgradeable makes it less efficient and unsuitable for long term usage.
2.4	system should have proper encryption when transporting data	laws/ legislation/ guidelines (see 11), risks (see 13)	The system needs to be secure, especially when handling data as per GDPR and to sustain trust with customers that their information is being stored securely.

10. Key Performance Indicators (KPI)

no.	Criteria	How generated	Why important
1.1	Rise in sales	Reviewing sales data	Seeing a rise in sales is crucial in rating the success of a new system because it shows that it is profitable.
1.2	Increased User Account creation	checking audit logs, seeing more accounts being added	An increase in account creation shows interest and use in the system by the consumer.
1.3	increased orders	reviewing sales data, checking audit logs, stock decreasing at a faster rate	An increase in orders means an increase in sales, which proves that the system is profitable
1.4	increased customer referrals	referral programs, KPI 1.2	customers referring each other to use the system proves that it is easy to use and like by consumers
1.5	increased amount of repeat visitors/ customers	audit logs, KPI 1.3	customers returning to view and order more products. Shows trust and loyalty with the company and system being built.

1.6	less security issues/breaches	risks (see 13)	Shows the system is up to date and secure to reduce the chances of a breach into the system. Keeping data secure as GDPR states (see 11)
1.7	increase in staff morale	staff feedback, amount of work done, staff retention	staff's happiness is crucial to a business. Happy staff work more efficiently and quickly to a high quality. It also improves staff retention as unhappy staff leave to work elsewhere.
1.8	loyalty rewards being claimed and used	reviewing loyalty scheme data, user stories (see 7)	People claiming their loyalty rewards proves that they are using the system and are ordering repeatedly. This shows the system keeps customers engaged and loyal to the service.

These all help boost businesses, the customers and staff liking and understanding the system can make or break the success of its implementation. After the system is built, GLH should see an increase in the figures stated above.

11. Laws/ Guidance/ Legislation

Data protection act 2018 / GDPR

The systems back-end database will hold sensitive data on all the users registered to the system (eg contact details, address, payment details etc). The data protection act ensures that companies are legally required to keep that data up to date, secure, deleted on request and only collect necessary data that is needed to operate. This means that companies need to consider this when developing a system.

Ensuring that they are using up to date security protocols to protect the system against attacks. Preventing them and ensuring the safety of data in the event that the system is breached.

Users need to be able to see all the data the system contains about them on request and have the right to have that data removed on request (the right to be forgotten).

Users also have the right to know what their data is being used for, this is usually disclosed in a user/ data agreement. Those documents need to state why data is being stored and what is used for. Data cannot be stored without a proper purpose and proper disclosure to the data subject.

For this project, writing a user/ data use agreement is outside of my scope for this project (see 4). In a real life scenario, someone with legal knowledge (like a lawyer) would be hired or locally used to draft up these documents to ensure that users are aware on what they are giving and how it is being used.

The Consumer Rights Act 2015

Under the consumer rights act. All goods sold online or in person have to be of satisfactory quality, fit for purpose and as described in the listing.

Consumers also have the right to refund, repair or replacement if the goods are faulty in anyway. Typically being within 30 days to return based on the product, company policy etc.

The Consumer rights act means that all product listings on the system need to be truthful, transparent and properly describe what the consumer is receiving (eg size of a cut of meat, grams (g)/ kilograms (kg) of the product etc) when ordering a product.

Equality Act 2010

The equality act 2010 ensures that all people are treated equally and fairly no matter age, disability, sex, gender identity etc. It criminalises discriminating against marginalised groups (predefined protected characteristics).

This law extends to the internet, meaning that websites and systems need to be developed to inclusive of all users and cannot be programmed to discriminate against them.

This is especially important on the user interface (what users will interact with the system with). The best way to accomplish this is by following the WCAG guidelines (see in detail below) and using other tools (eg generating a lighthouse report using google's developer tools) to ensure that the website is accessible to all users.

Using tools and guidelines is the best way to do this as they are already curated and developed to consider all users so you won't unintentionally miss anyone out.

Web Content Accessibility Guidelines (WCAG)

WCAG guidelines tell developers what they need to include whilst designing and developing systems to ensure that it is inclusive to all users without bias.

WCAG is designed to accommodate for everyone and whatever accessibility tools they may be using to browse the internet (eg screen readers, mouse trackers etc).

The guidelines are being kept up to date and updated to meet changing needs and accessible technologies being developed.

12. Alternative Systems

The Dorset Meat Company

What does the system do well?

- The system is very straight forward and shows all their products off easily and on the go. You can see all the different products they have for sale on their nav bar, a clear way to see your basket and how to login.



BUY MEAT ▾ EASTER MEAT BOXES ▾ BBQ ▾ FISH ▾ COOK'S KITCHEN CHEF PREPARED ▾ CHEESE ▾ DELI ▾ NEW GIFTS ▾ ABOUT ▾

- The system has clear indication on where to find out where the products are sourced on the home page. This shows that they are clear about how their food is sourced and the people buying can know where it comes from.



MEET OUR DORSET & WILTSHIRE FARMERS

We all want to know where our food comes from. We all want to have confidence in the provenance and integrity of the food we are eating.

We work exclusively with small, family-run farms in Dorset and Wiltshire that all produce ethically reared, nutritious meat, in a natural growing environment, in harmony with nature. Animals that have been allowed to grow at a natural pace and have not been rushed.

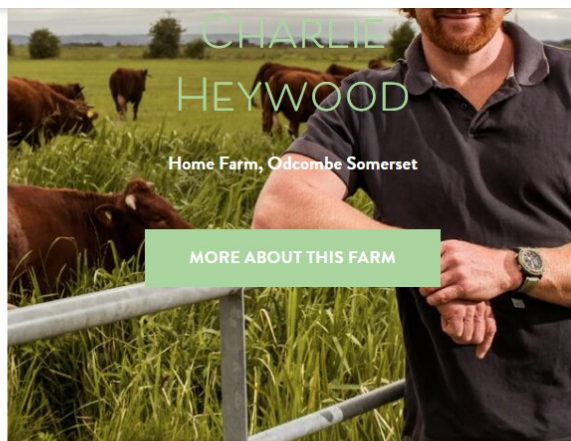
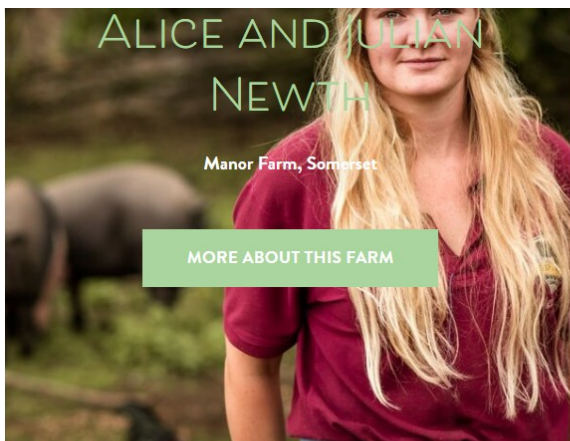
By shopping with The Dorset Meat Company, you are supporting small, family-run farms that have been an essential part of our rural economy down the ages.

SEE OUR FARMS



t 10% Off

- When you click on the link, you can learn about each individual farm and the people running it. Which again shows exactly where and who the products are coming from.



- # LOGIN

Username or email address *

Password *

☐ Remember me

LOG IN

[Lost your password?](#)

Use a social account for faster login or registration.



Log in with Google



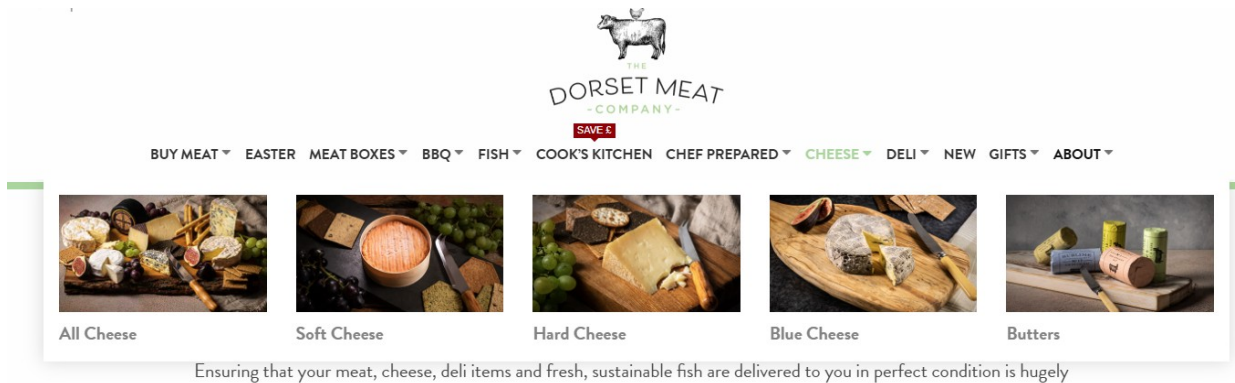
Log in with Facebook

NEW TO THE DORSET MEAT COMPANY?

CREATE AN ACCOUNT

- Their drop down menus are easy to navigate and click and don't disappear straight away when clicking off them which gives the user time to get to where they want to select and gives them a chance to go back to where they were if they accidentally misclick off of it.

- Some of the smaller sections also display pictures which is really nice visually on the eyes, makes the webpage look more professional and put together. As well as clearly showing what a user is looking for.



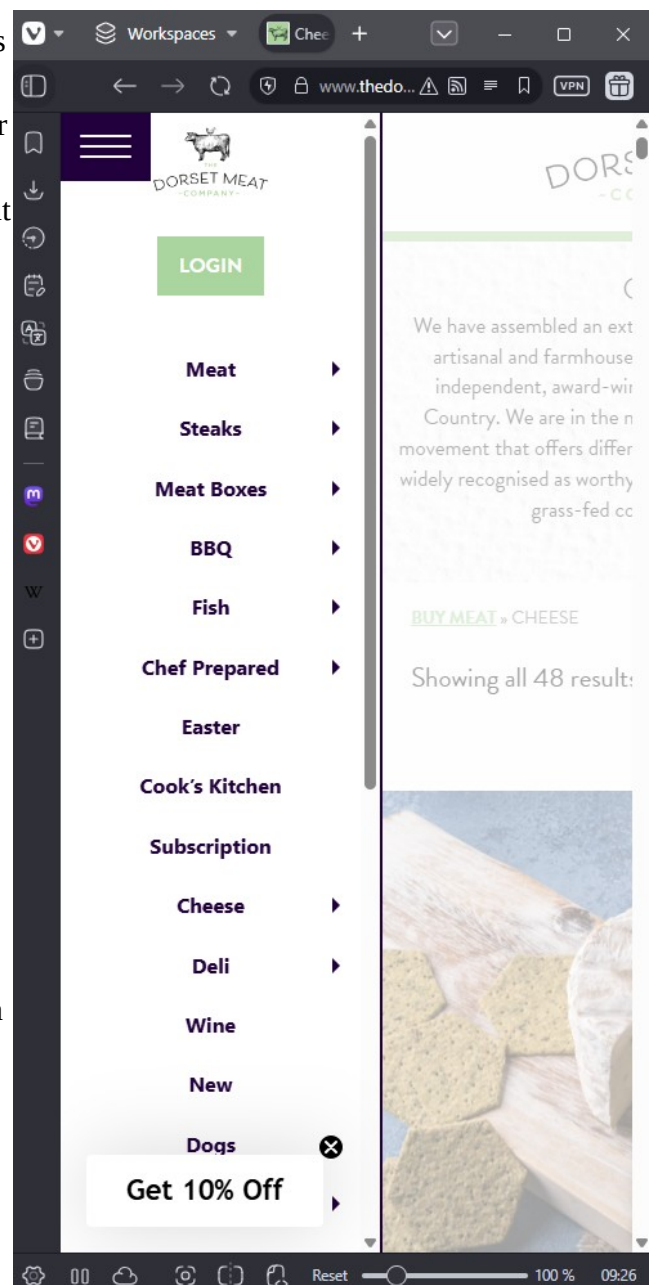
What does the system fail to do?

- The system only shows the option to sign up through the login page which may confuse users who look for a sign up tab and can't find one. The login page is also hidden away in the corner where users don't tend to look straight away when it could be on the nav bar next to the about tab.
- The webpage also glitches when you resize it and the smaller mobile layout sizes aren't as thought out as the full screen view which is an oversight because most users will be using mobile devices.
- It is quite squished and shows some different options which you can't see on the full screen view like things for dogs and subscriptions.

What will you take forward into your design and why?

I will take forward their clear layout into my design as the buttons are large and easy to click on. All the information is clearly laid out and doesn't feel like it's hiding away from you. So it's easy to access so you can learn more about how the product is sourced.

The layout is typical with shopping sites which also makes it easy to understand coming onto it for the first time.



Blacker Hall Farm

What does the system do well?

- When you open the webpage, it shows the family that runs the farm which immediately gives the site a more wholesome and local feel to it.



- They show all the different things they offer straight up and clearly. Showing their shopping section, visiting section etc on the nav bar and on the home page of the website. Showing how to learn about the farm, and you can easily see all the different things they offer.

BLACKER HALL FARM
GREAT YORKSHIRE FRESH FOOD

HOME SHOP ONLINE ▾ VISIT ▾ EAT & DRINK ▾ LITTLE FARMERS ▾ EVENTS ▾ ABOUT ▾ CONTACT ▾

- They Clearly display seasonal event and workshops they are offering on the front page so you can clearly see what is on so you can book onto it. This improves the appearance of the website as well as promoting what they have to offer.



Mother's Day dining in The Barn Kitchen
12th - 15th March

Book your table today for our extra special Mother's Day Roast Dinner or Afternoon Tea in The Barn Kitchen.

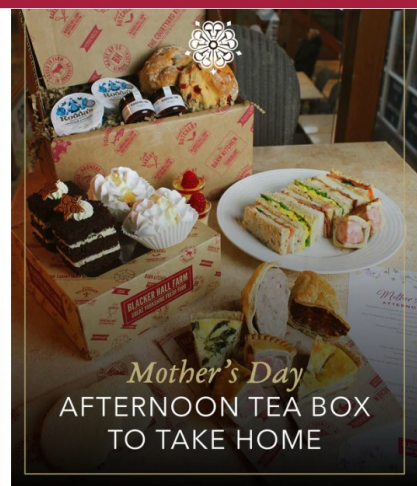
[Book your table today >](#)



Lamb & Kid(Goats) Bottle Feeding
NOW OPEN

A truly unique, up close and personal animal experience for the family at our Little Farmer Trail.

[Book your tickets today >](#)



Afternoon Tea to Take Home

Available Fri 13th - Sun 15th March

Treat Mum to something truly special this Mother's Day with our beautifully prepared **Mother's Day Afternoon TeaBox to take home**

[Pre order yours today >](#)

What does the system fail to do?

- On shopping pages with an image background, they choose to use black plain text which is harder to see with the background compared to if they gave the text a background to separate it or make the text larger and bolder.



- They also fail to show any deals that are available or login. This means the user may not have the functionality to sign-up and login to any reward schemes and doesn't allow repeat customers to easily view their order history or set up any regular shipments of products to their business or household.

What will you take forward into your design and why?

I will take the simple family feel into my design as it pushes that the products are made local by normal working people and not large corporations who don't care about you. It makes you feel like

you're looking for products from your friends or someone you could be close too instead of a mysterious corporations who hides where products come from and who produces it.

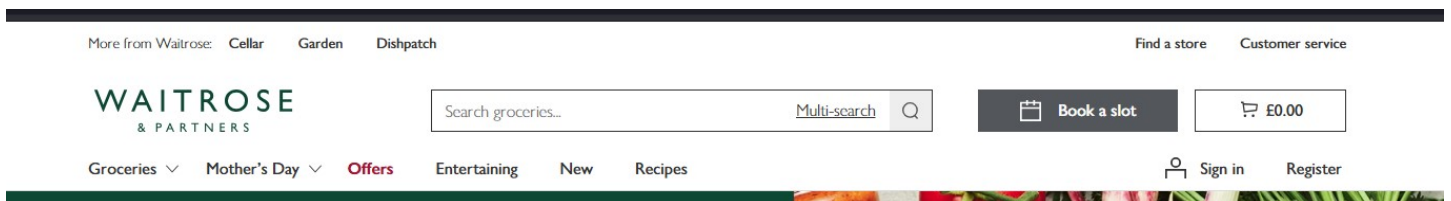
Waitrose & Partners

What does the system do well?

- The system clearly displays login and registering on the nav bar and clearly organises all the things they offer into sections on the nav bar.

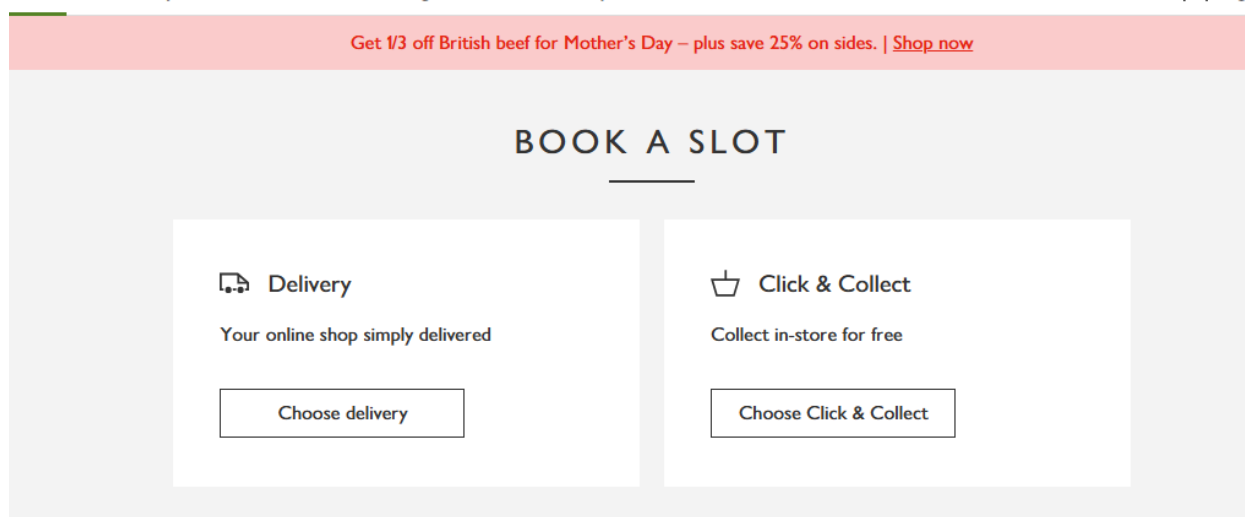
It clearly lets the user see where they can book a slot for delivery or collection in a separate colour and shows the basket and running total for when users are shopping.

The sign in and register tabs are kept separate from the food itself ,to reduce the about of information in one place, whilst keeping it visible. Where the user will see it.



- When going to book a slot it clearly allows you to select to collect it in store or have it delivered to your home. This means that users aren't accidentally selecting the wrong thing and the orders can be kept separate to reduce mixing up orders.

It also clearly shows seasonal sales they have on currently with an option for a user to select to be taken to it. This promotes customers to purchase products they may not of considered because it on sale and keeps it transparent on what is on offer.



What does the system fail to do?

- The system fails to allow the user to login with an api, forcing the user to have to make an account. This can discourage users from using the service because they don't want to fuss

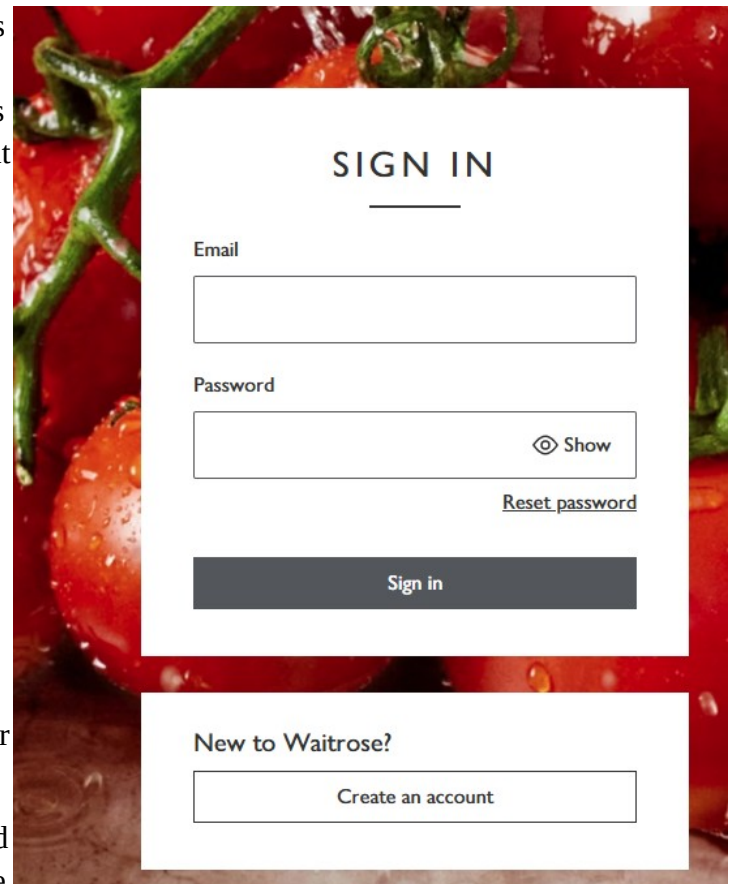
about, signing up and filling in information they already have stored in an API of their choice (e.g. google, apple etc)

- The system also fails to show where products are sourced so customers won't know where their products have come from. This removes the personal local aspect. Increase the amount of air miles in consumers food and damages local economies because food is being flown in from other countries and profits are being sent to big countries and CEOs rather than local farmers and people.

What will you take forward into your design and why?

I will take forward keeping the register and login on the system visible to the user and clearly marked for each one. To prevent confusion and allowing the user to make an account.

I will also take forward, separating the collection and delivery options to make separating the orders on the staff interface easier to reduce mistakes made.



SIGN IN

Email

Password

Show

[Reset password](#)

Sign in

New to Waitrose?

Create an account

13. Risks

External Risks and delays

- Rough weather conditions or cases of natural disaster can cause power outages, inability for staff to travel to work, destruction of where data is stored etc. This can cause the project to be delayed because developers can't work on the project if they can't access their work or work is lost. These weather conditions can be predicted so it is important to keep up to date with the weather in the local area and ensuring that protocols are in place to protect the staff and data. (eg emergency power sources, remote work etc)
- Data breaches can cause for data to be stolen (held for ransom, ransomware) or deleted which could cause for work to be lost and delay projects. It is important that during development that data is stored securely and backups are kept. It is also important that all testing data is fake data that is fine to lose, rather than actual data about real people.
- Changes in legislation can cause for change in office environment and how code has to work (eg new law requiring social media to have age verification) can cause for the system to

have to have features implemented, changes in how it being developed and at an extreme causing for the project to have be started over.

To mitigate this, it is important to keep up to date with current legislation and ensure that the finished system is adaptable to change and changing needs so that required features can be altered, added or removed to keep up to date with the law.

Resource and skill risks

- Staff Leaving the company in pursuit of new opportunities during the project can mean that someone who has worked on the system and understand its is no longer there to give insight and help others and new staff who will be taking their place. This can delay the project as time will have to be taken to fill their role, train up new staff and have them understand the code.

This can be mitigated by ensuring that multiple people are working on the system, understanding the ins and outs of it. As well as proper commenting throughout the code so that new staff looking in at it will understand it.

- A lack of skill developing the system to how the client wants it can delay development of a system as staff will have to spend time learning how to make the system meet what the client wants it to do.

This can be mitigated by keeping up to date with current security protocols, ui styles etc to stay with the times of what the client wants.

Ensuring staff work on areas they are most skilled at (as mentioned in 3, scope) time is saved and there will be less issues when it comes to skill.

Other risks

- Changes in client needs. The client changing their mind of certain features, website appearance etc can delay the project because features may have to change how things are done to meet what the client wants.

To mitigate this, it is important to keep clear communication on how things are being done, progress being made etc with the rest of the dev team and the client. This not only gives the client peace of mind, but can mean that changes can be made before the system is completely finished. As well as in the case that the project does have to be delayed, there is a good relationship built up with the client and the client can understand why and how long it will take to make the decision on what to do.

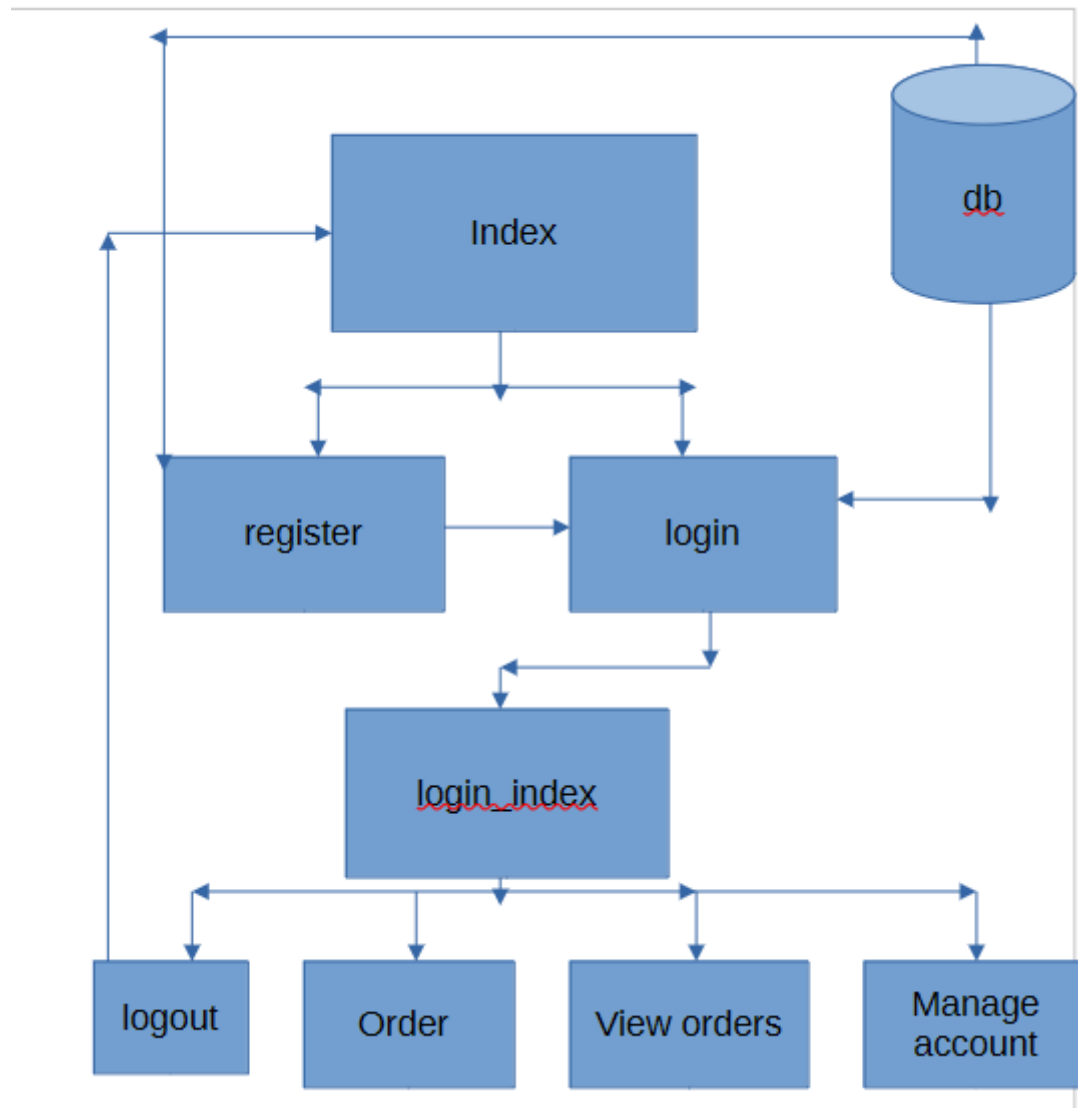
- Security risks. Risks in the security of the system can cause data to be stolen/ intercepted and puts the whole system at risk of being hacked as well as breaching the law (eg GDPR, see 11).

To mitigate this, staff need to keep up to date and trained on implementing current security protocols into the system, (up to date database connections, encryption methods etc). To ensure that they are using modern techniques that are designed against latest hacking and infiltration methods.

Rigour and precise testing can ensure that all back doors can be sealed and kept secure. To prevent entry.

Protecting the data itself is the most important thing so ensuring that in the case of someone breaching the system, encrypting and preventing them from, taking and altering data is crucial.

14. Decomposition



To order products, view past and present orders and receive rewards via the loyalty scheme, users will have to make and login to an account to be able to log those details in the system.

The system will need a database to store details on products, users etc. This is crucial to ensure that data on users is correct and stored securely.

For clarity on my decomposition diagram, I chose to not add arrows indicating where the database will be interacting with each unit of the system. In reality, the database will required anywhere where user, product and order data can be seen. Eg, sign up, login, product pages, order history etc.

Every unit will have different pages to allow users to successfully manage their account and order products, whilst keeping the system clear and easy to use (NFR 1.1, FR 3.1).

Further details will be shown in the design document.

15. Description of Proposed Solution

For the Solution, Greenfield Local Hub (GLH) will need a professionally built e-commerce website to sell products online for delivery/ collection to reach the target demographic of adults who need to purchase products for their household/ business etc that is locally sourced from farmers, and informs the consumer where the product is from, what it contains etc. This ensures that product listings are kept transparent and in accordance of the consumer rights act (see 11).

general overview

I will build an e-commerce system that allows users to select/ add products to a digital shopping basket, purchase products for delivery or collection, alter orders for a set period of time before the delivery date with a loyalty scheme and rewards (discounts) that will be offered to the user in the account dashboard.

The system will be compliant with law/ legislation/ guidelines to ensure the website is accessible to all users (WCAG, Equality act 2010), has up to date and proper security protocols to prevent malicious breaches of the system and to protect data (GDPR, data protection act 2018) and encourages users to recommend the system to others and order repeatedly.

With the system, ensuring that local farmers and suppliers are known and advertised to the user is of mass importance. This means that I will include information pages on the farmers/ suppliers and products will be marked with what farm. Supplier they have come from.

Account system

For the account system I will make a system that allows users to register and login to the system, view, update and change account details and receive offers and rewards under a loyalty scheme program. This will be accomplished by offering rewards based on how many orders a user has made and based on analytics on what a user has purchased.

Under the scope of my project (see 4) I will not be including a 2FA on login (verifying with email, phone etc) or logging in using an API (google, apple etc). In a real development scenario, I would include one to allow users an easier way to login to the increasing the amount of accounts created (KPI 1.2).

Order system

For the order system of my project, I will build a system that allows users to add items to a basket at their chosen quantity and then checkout the basket for either collection or delivery to meet GLH's and the customers needs.

All Products will be clear and easy to understand for users and adhere to food regulations (see 1) displaying what products are, how much you will purchase and where they have come from.

Under the scope of my project (see 4) I will not be handling any of the financial data required for the system to function as a proper e-commerce site.